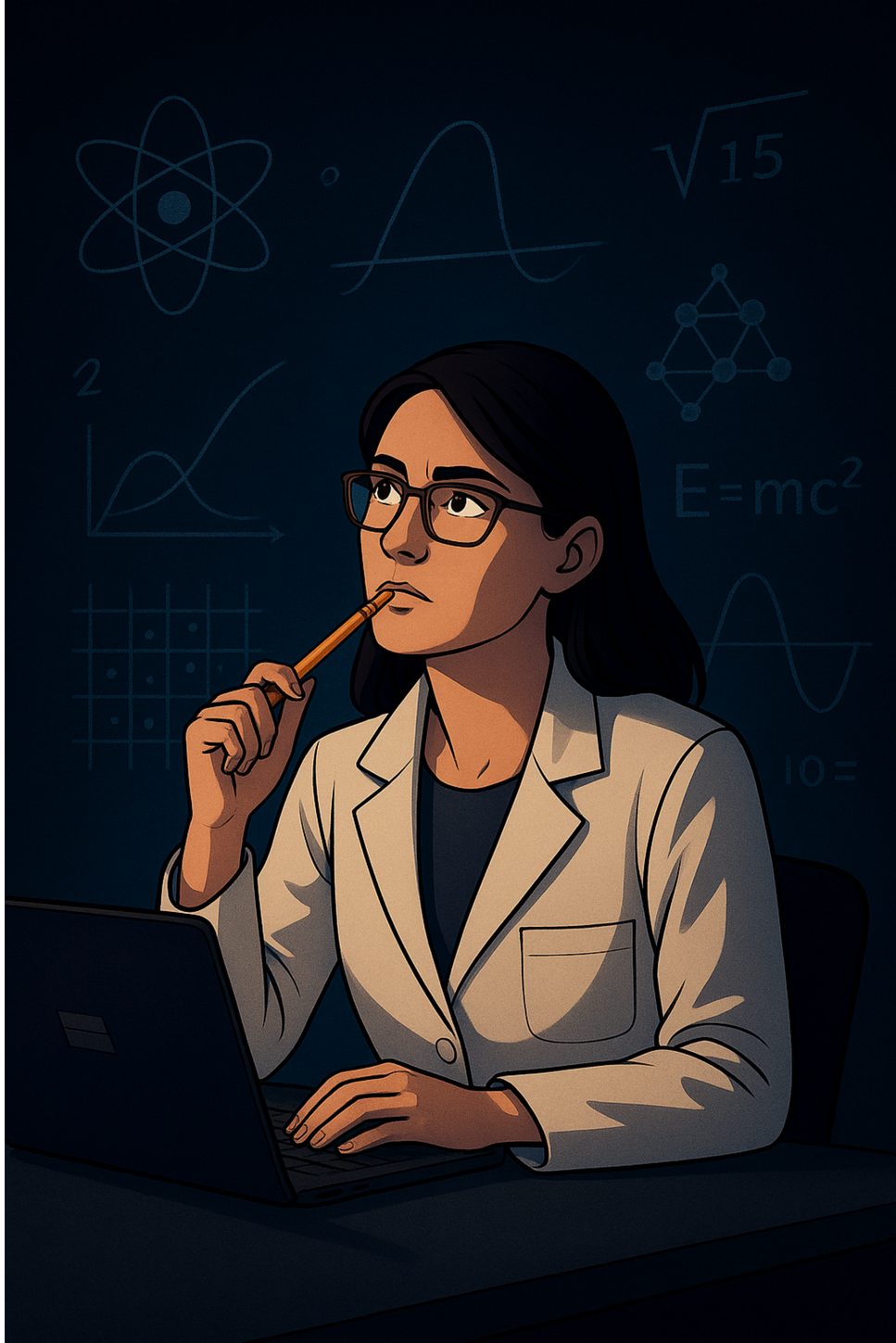
Statistics and Data Analysis

Week 4

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UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA



Program of the first 4 weeks (March)

Days	Thu 12:45 - 14:30 (2h)	Fri 13:45 - 15:30 (2h)
05/03 06/03	- Probability: foundations and interpretations - Practice with PDFs, CDFs, Quantiles - Joint PDFs, moments, correlation, transformations	- Practical CLT and the lognormal - Linear error propagation - MC methods - **Exercise**
12/03 13/03	- Frequentist hypothesis testing - Optimal tests, Neyman-Pearson, p-value - Estimators and properties	- Likelihood - MLE - **Exercise**
19/03 20/03	- Asymptotic MLE properties and MVB - Advanced MLE applications and MC methods - Frequentist Confidence intervals	- Profile likelihood and Wilks' theorem - **Exercise**
26/03 27/03	- Bayesian priors and posterior construction - Bayesian parameter estimation	- Bayesian hypothesis testing - **Exercise**



Jupyter notebook

Week3_Exercise_Solutions.ipynb

Bayesian parameter estimation

Bayesian philosophy

- Bayesian statistics can assign a probability to a hypothesis, but requires a prior (subjective) probability on the hypothesis:

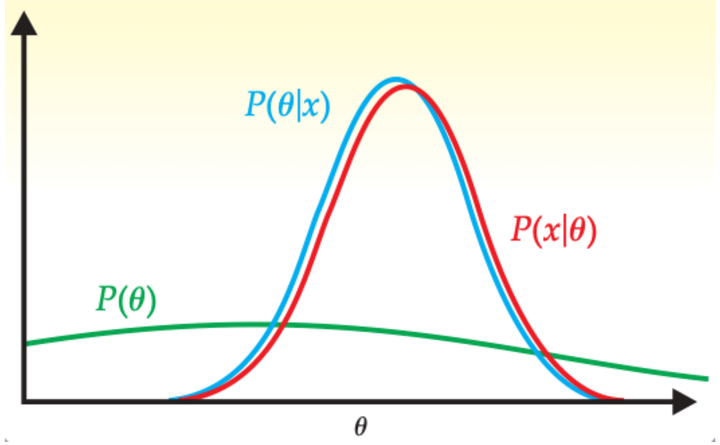
probability of the data assuming hypothesis H (the likelihood)

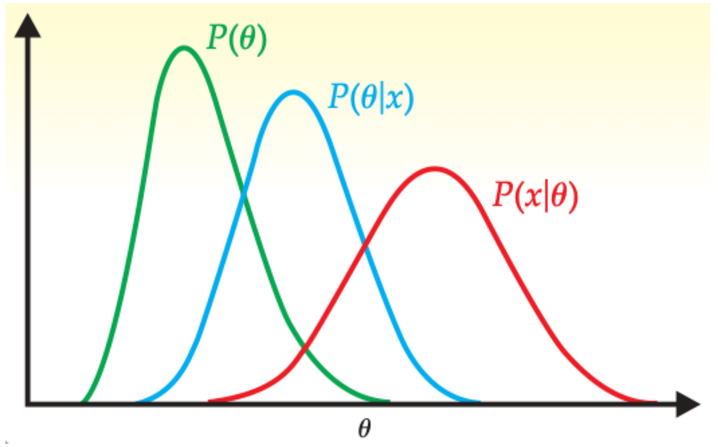
prior probability, i.e., before seeing the data

$$P(H|\vec{x}) = \frac{P(\vec{x}|H)\pi(H)}{\int P(\vec{x}|H)\pi(H)\,dH}$$

posterior probability, i.e., after seeing the data

normalization involves sum over all possible hypotheses

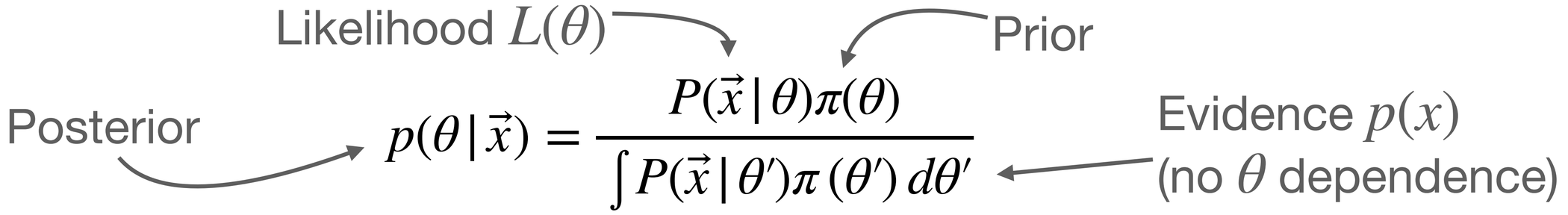




- If you prior probability of H is $\pi(H)$, then in light of the data you can update the probability to $P(H|\vec{x})$
- However: no general prescription for prior probability!

Bayesian parameter estimation

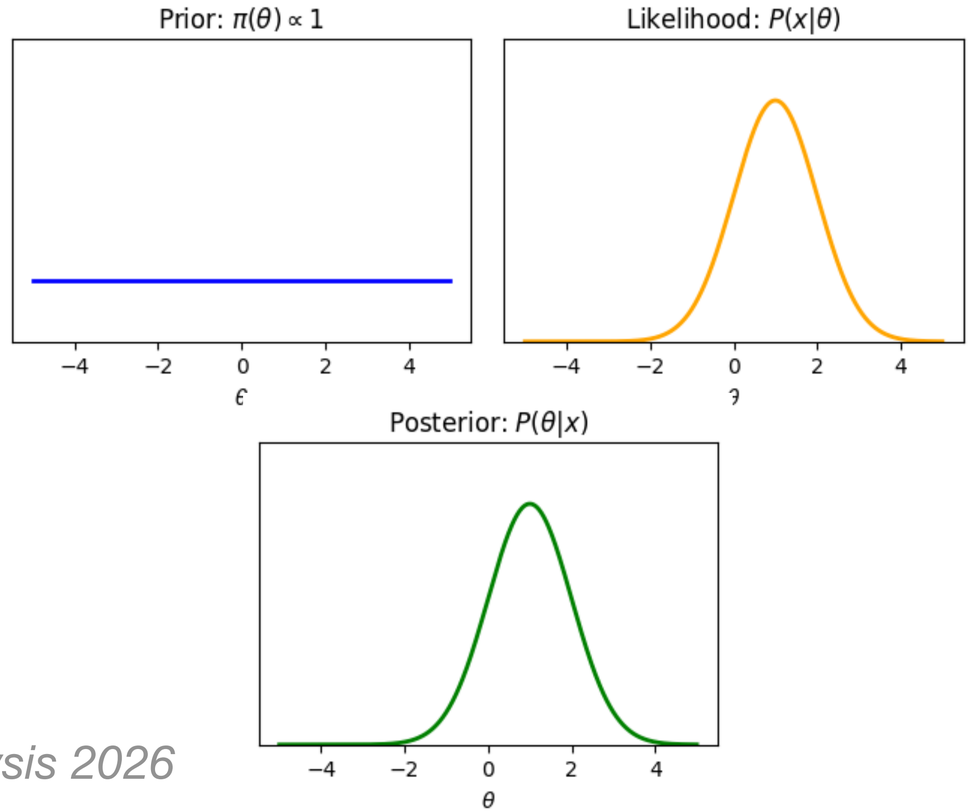
- The parameters θ are random variables and have probabilities
- Probability reflects the degree of belief that θ takes a certain value
- Before doing the experiment you need a prior PDF for $\pi(\theta)$
- Then use Bayes' theorem to update the prior in the light of data



- The posterior $p(\theta|\vec{x})$ contains all the information about θ!

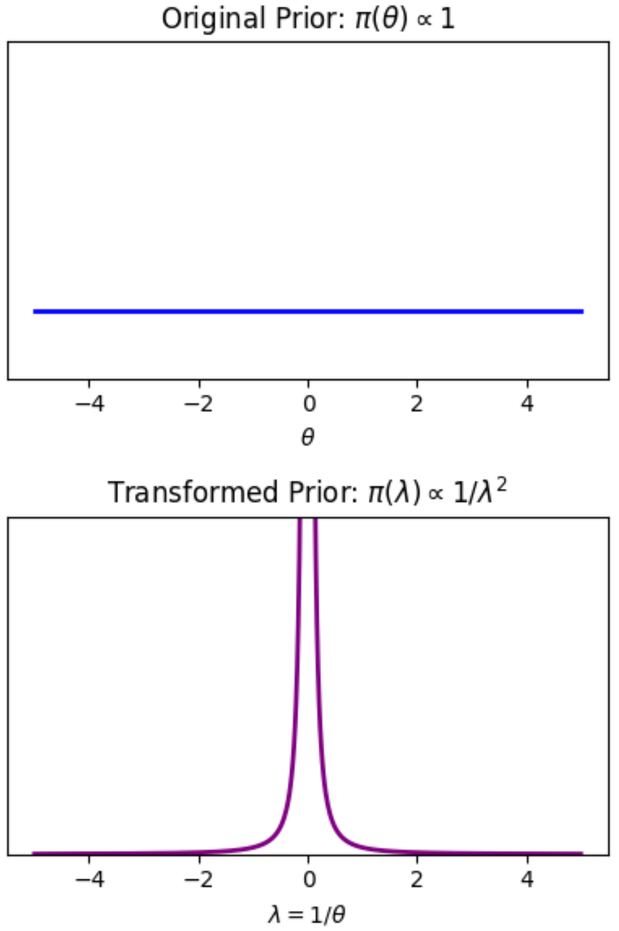
Posterior summary: **Point Estimate**

- The pragmatic Bayesian might want a point estimate of θ that summarizes the posterior PDF
- No clear rule: mode, median or exp.value of the posterior
  - Your choice might depend on the shape of the posterior but is arbitrary
- One special case:
  - Assume a flat prior $\pi(\theta) = \text{const}$
  - Then $p(\theta \mid x) \propto P(x \mid \theta) = L(\theta)$
  - If you take the mode of the posterior (also called MAP = Maximum A Posteriori) $\Rightarrow \hat{\theta}_{\text{MAP}} = \hat{\theta}_{\text{MLE}}$



Prior ignorance

- One can use $\pi(\theta) = \text{const}$ in some cases
  - Note that it's not normalised (improper prior)
  - Still allowed as long as the posterior can be normalizable
- However, $\pi(\theta) = \text{const}$ does not represent "prior ignorance"
  - One could for example reparametrise the problem with $\lambda = 1/\theta$
  - Then $\pi(\lambda)$ would not be flat: $\pi_\lambda(\lambda) = \pi_\theta(\theta) \left| \frac{d\theta}{d\lambda} \right| \propto \frac{1}{\lambda^2}$
- Complete prior ignorance is not well defined
  - You have to assume a prior PDF to get the posterior



Choice of prior

- **Informative priors**
  - The prior pdf could be derived from the posterior pdf of previous experiments
  - The shape of the priors could be derived from physical principles, e.g. the signal rate parameter should not be negative, efficiency parameter $\in [0,1]$
- **Uninformative priors**
  - Building a completely “agnostic” prior is often impossible
  - It could reflect the symmetry of your problem: e.g. translation invariant (uniform) or scale invariant ($\pi(\theta) \propto 1/\theta$ so that it’s uniform for $\ln \theta$)
  - Several sophisticated prescriptions in the literature:
    - Conjugate priors, maximum entropy principle, Jeffrey’s prior
    - If you are interested, read Pruneau’s section 7.3 Chosing a Prior

Sensitivity

- Always check if the choice of “uninformative” prior has a significant effect on the posterior distribution
- **Sensitivity** of the posterior on the choice of prior requires care!
- As long as your scientific result is given in terms of “if this prior then this posterior” and the likelihood is provided, then your result is sound
- Providing the likelihood is generally good practice independently of whether you are Bayesian or Frequentist in your interpretation of probability



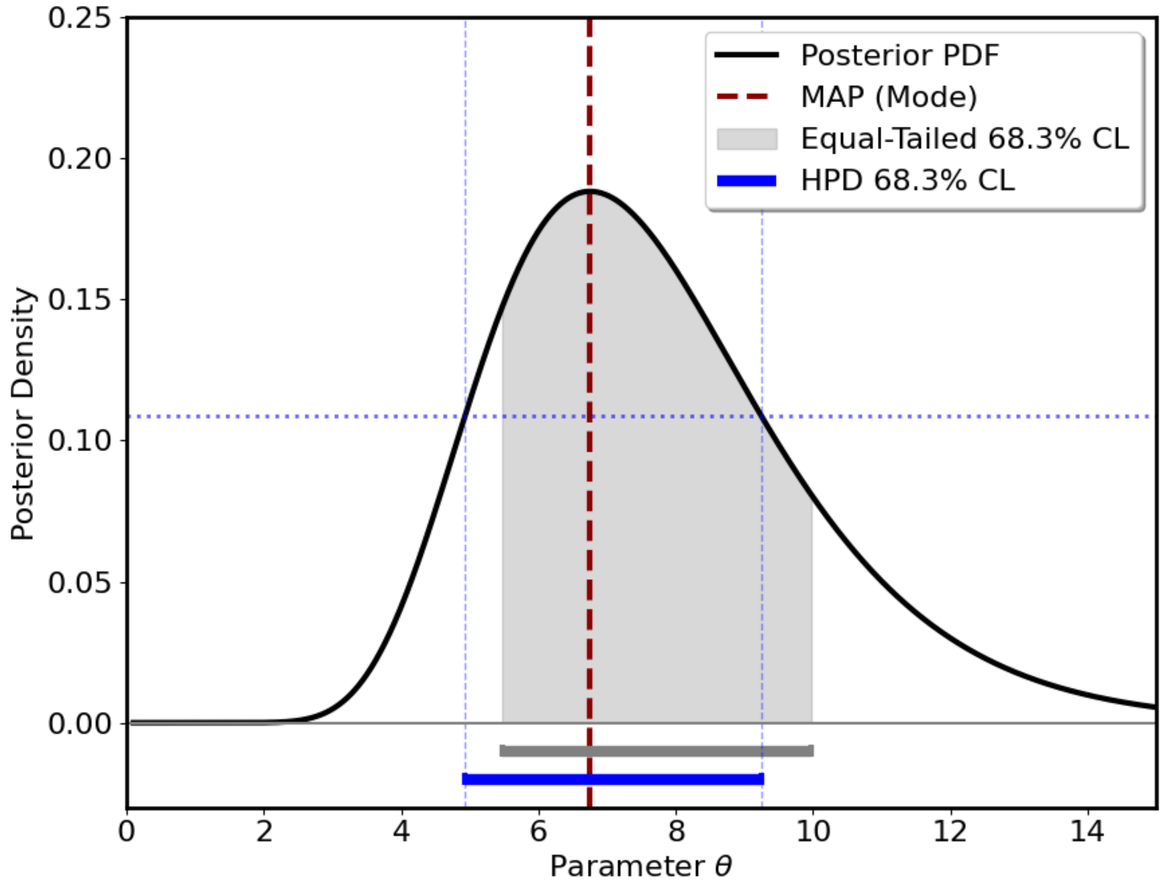
Jupyter notebook

Week4_BayesianInference.ipynb

Bayesian credible intervals/regions

Posterior summary: **Credible Intervals**

- One can also define intervals that summarise the posterior
- As Bayesians we will call them **credible intervals** at a certain CL (Credible Level)
- Two ways of defining them:
  - **Equally tailed intervals** (leave same fraction of pdf on left and right tail)
  - Region of **Highest Posterior Density** (HPD)
  - For both, the size is simply defined by the posterior integral within the interval matching the CL



Meaning of credible intervals

- A Bayesian credible interval makes the exact statement that almost everyone incorrectly assumes a frequentist interval makes (e.g. for 95% CL):

  - **Bayesian credible interval:** given the data we just observed (and our prior knowledge), there is a 95% probability that the true parameter lies within this specific interval.

  - **Frequentist confidence interval**: if we repeat the experiment an infinite number of times, 95% of the obtained intervals contain the true parameter value

Bayesian philosophy:

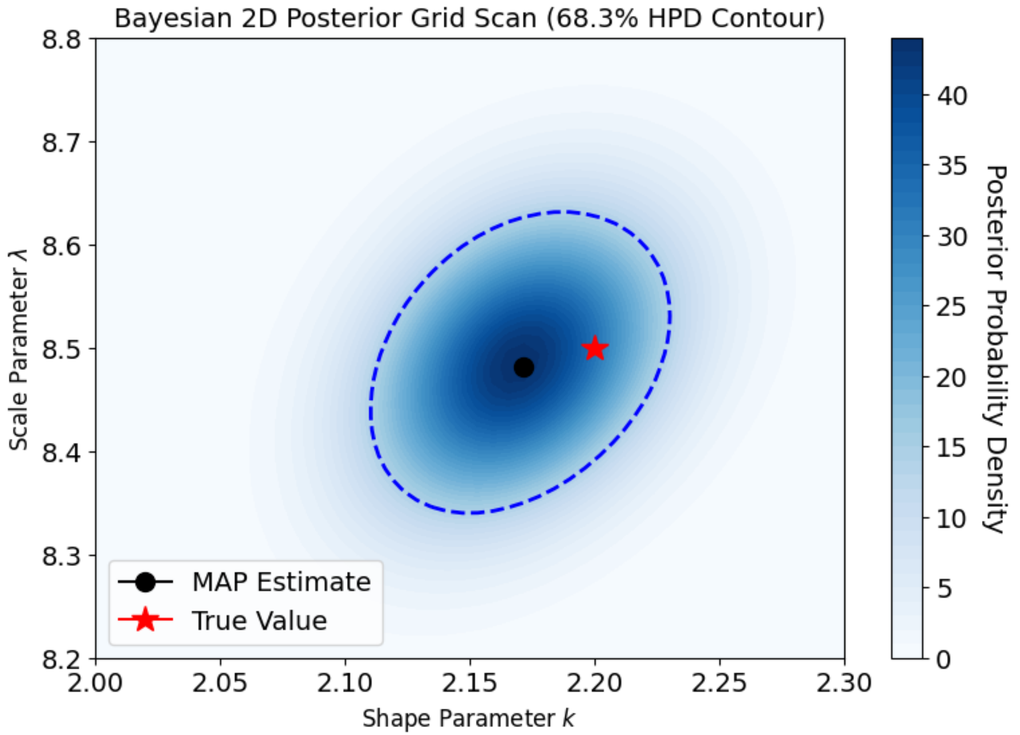
- The dataset is fixed (there's only one)
- The parameter is a random variable to which we assign a probability

Frequentist philosophy:

- The dataset is the random variable and probability is defined in the infinite repetition of the experiment
- The parameter is fixed

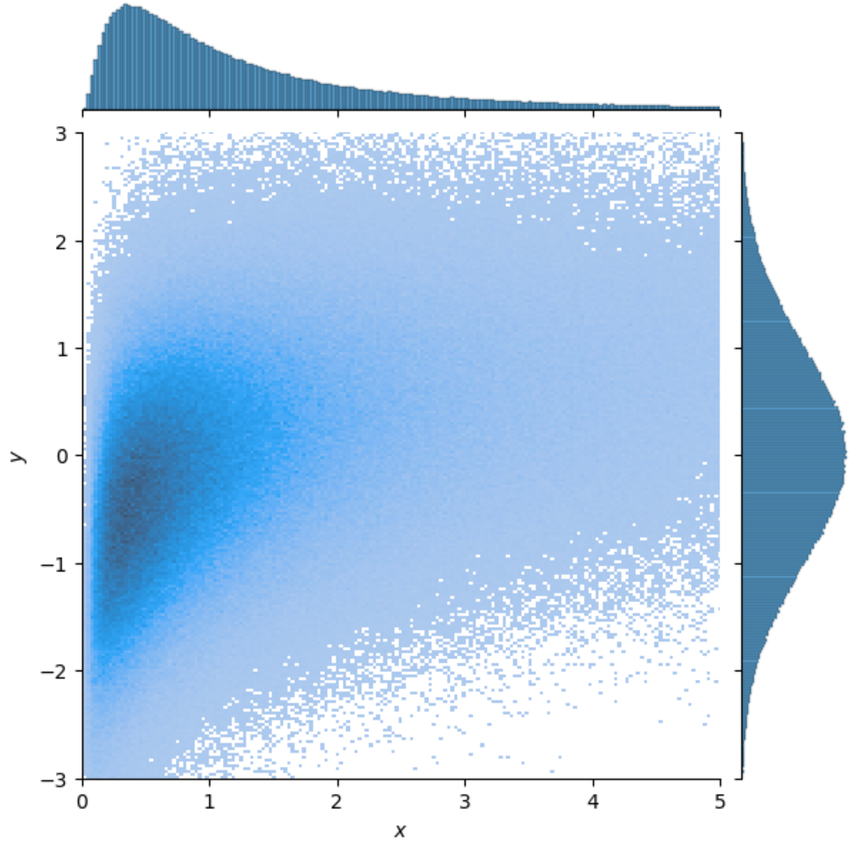
Posterior summary: **Credible regions**

- For several parameters credible intervals → **credible regions**
- No ordering of parameter values → can't define "equally tailed regions"
- The standard is to use **HPD** (Highest Posterior Density) to define credible regions



POIs and nuisances

- In Bayesian statistics it is much easier to get the PDF for your parameters of interest (POIs) while eliminating over the dependence on nuisance parameters
- It is sufficient to marginalise the PDF over the nuisance parameters:
- Posterior PDF for POIs $\boldsymbol{\mu}$: $p(\boldsymbol{\mu}) = \int f(\boldsymbol{x}, \boldsymbol{\nu})\, d\boldsymbol{\nu}$
- No need for likelihood ratios and Wilks' theorem. The posterior PDF contains all the information.







Jupyter notebook

Week4_CredibleRegions.ipynb

Markov Chain Monte Carlo

- Calculating the posterior with many parameters is very hard with a grid scan or accept-reject MC as they scale exponentially (a 100-point grid with 5 parameters involves 10^{10} points)
- Solution: **Markov chain Monte Carlo (MCMC)**
  - MCMC will generate a **correlated** sequence of random numbers that samples preferentially the regions of high probability density
  - MCMC will ignore the vast empty regions of parameter space
  - Runtime of MCMC scales roughly linearly with dimensions!
  - Several different algorithms with pros and cons
- **MCMC has revolutionised Bayesian computations**

Example: Metropolis-Hastings MCMC

- **Goal**: given an target n-dimensional pdf $p(\vec{\theta})$ generate a sequence of $\vec{\theta}$

1. Start at some point $\vec{\theta}_0$
2. Generate a point $\vec{\theta}$ from a symmetric pdf $q(\vec{\theta}; \vec{\theta}_0)$
   (e.g. gaussian centered around $\vec{\theta}_0$)
3. If $p(\vec{\theta}) > p(\vec{\theta}_0)$ then take the proposed point $\vec{\theta}_1 = \vec{\theta}$

   else, generate u from uniform in $[0,1]$ and take it only if $u \leq \frac{p(\vec{\theta})}{p(\vec{\theta}_0)}$
4. If you don't take the proposed point, repeat old point $\vec{\theta}_1 = \vec{\theta}_0$
5. Iterate

Random walk with Metropolis-Hastings MCMC

animation from chi-feng website

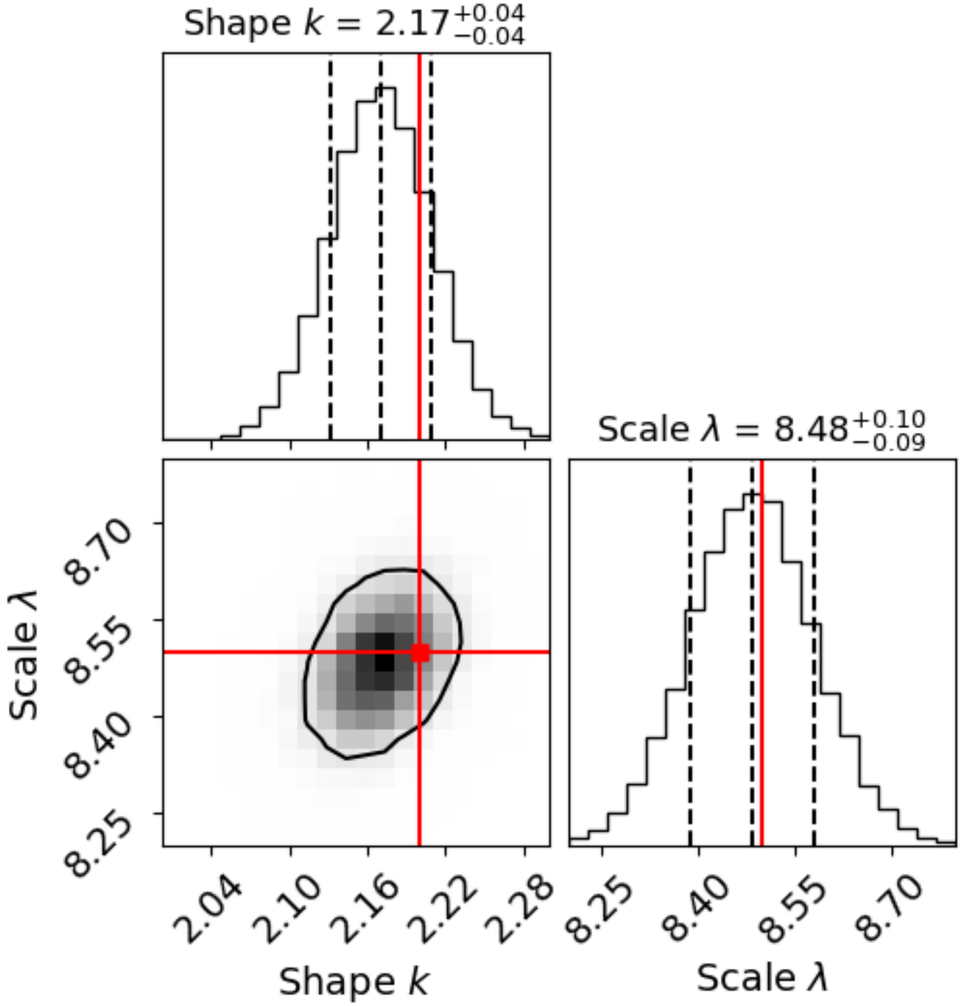
Common pitfalls of MCMC



- **The "Burn-in" phase**: sequences will take some time to reach the high-probability density region of the target pdf → remove first N steps
- **Autocorrelation:** each sample point i is correlated with $i+1$. Cannot be used as independent draws from normal MC sampling. Effectively, the statistical power of your MCMC sample is less than $\sqrt{N}$
- **Convergence**: sometimes the chain gets stuck in a local minimum and stays there while ignoring the global minimum → better to run multiple independent chains from different starting points.
- **Vast literature of tricks to get a stable MCMC sampling algorithm of complicated posteriors**

Typical Bayesian workflow

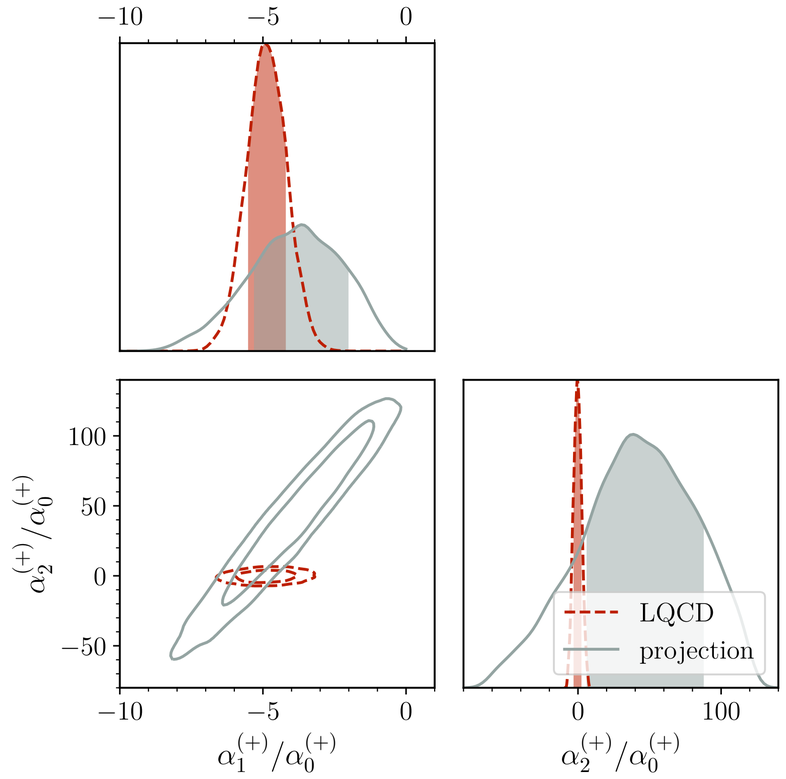
- Model specification
  - Define a prior $\pi(\theta)$ (or test a few)
  - Define the Likelihood $p(\mathrm{Data}\,|\,\theta)$
- Calculate the posterior using Bayes'
  - $p(\theta\,|\,\mathrm{Data}) \propto p(\mathrm{Data}\,|\,\theta)\pi(\theta)$
  - Use Markov Chain Monte Carlo (MCMC) to sample the posterior distribution
  - The posterior sample represents your posterior pdf
- Summarise the posterior pdf
  - Get marginalised posterior pdfs for your POIs
  - Define confidence intervals/regions



Examples from the literature

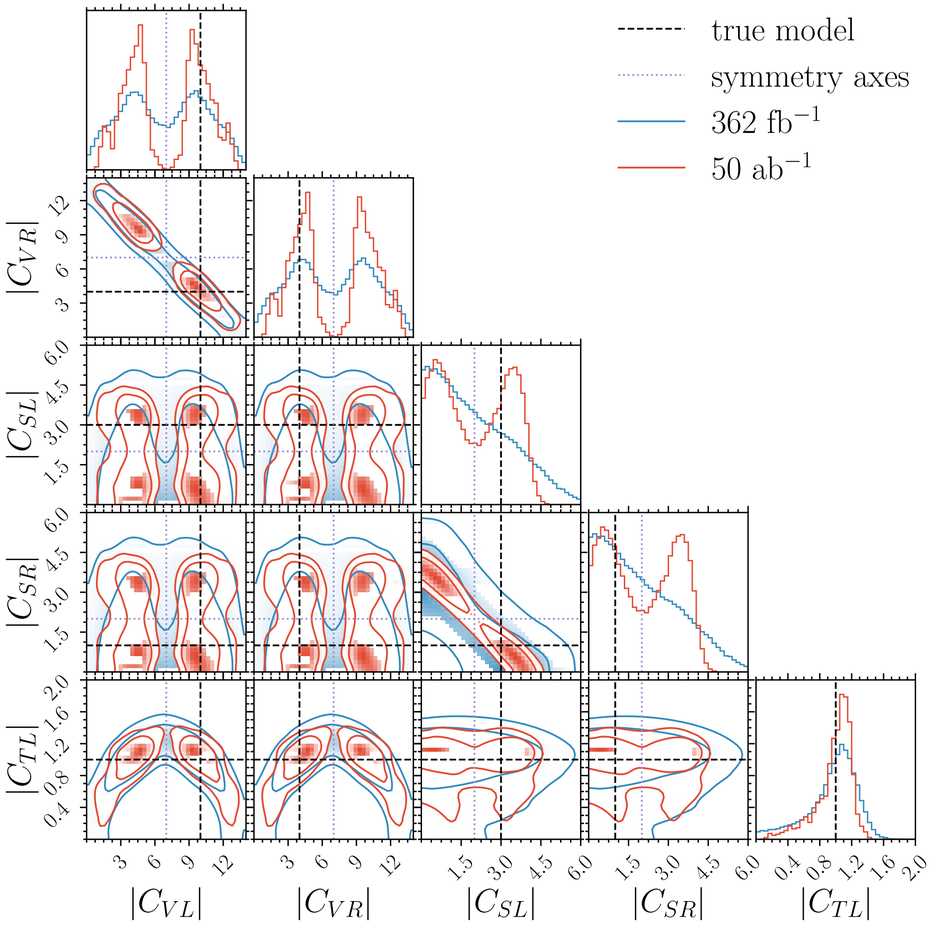
https://inspirehep.net/literature/2758034

Exploiting Perpendicular Momentum Distributions of Semileptonic Decays: $\bar{B}_s^0 \to D_s^+ \mu^- \bar{\nu}$ as a Case Study



https://inspirehep.net/literature/2758034

Constructing model-agnostic likelihoods, a method for the reinterpretation of particle physics results



Bayesian hypothesis testing

Hypothesis testing: Posterior odds

- The probability of a hypothesis given the data is simply the posterior
- Comparing different hypotheses is as simple as taking the ratio of the posteriors (often called posterior odds)
- Using Bayes' theorem the ratio becomes:

Posterior odds

$$\frac{P(H_1 \mid \mathbf{x})}{P(H_0 \mid \mathbf{x})} = \frac{P(\mathbf{x} \mid H_1)}{P(\mathbf{x} \mid H_0)} \frac{\pi(H_1)}{\pi(H_0)}$$

Note that factors $P(x)$ cancel in the ratio

Bayes factor

Prior odds

Bayes factor

- If you assign the same prior probability to the two hypotheses $\pi(H_1) = \pi(H_0)$
- Then the posterior odds is given simply by the Bayes Factor K

$$\frac{P(H_1 \mid \mathbf{x})}{P(H_0 \mid \mathbf{x})} = \frac{P(\mathbf{x} \mid H_1)}{P(\mathbf{x} \mid H_0)} \frac{\pi(H_1)}{\pi(H_0)} = \frac{P(\mathbf{x} \mid H_1)}{P(\mathbf{x} \mid H_0)} = K$$

- The Bayes factor can be thought as measuring the weight of evidence of the data in support of H_1 over H_0 and can be used similarly to a p-value
- A large K means that H_1 is preferred over H_0
- Jeffrey had a scale to interpret the Bayes factor in terms of “evidence”

Jeffrey’s scale for Bayes factors

Grade 0.	$K > 1$.	Null hypothesis supported.
Grade 1.	$1 > K > 10^{-1/2}$.	Evidence against q, but not worth more than a bare mention.
Grade 2.	$10^{-1/2} > K > 10^{-1}$.	Evidence against q substantial.
Grade 3.	$10^{-1} > K > 10^{-3/2}$.	Evidence against q strong.
Grade 4.	$10^{-3/2} > K > 10^{-2}$.	Evidence against q very strong.
Grade 5.	$10^{-2} > K$.	Evidence against q decisive.

Jeffreys, Harold (1998) [1961]. *The Theory of Probability* (3rd ed.). Oxford, England. p. 432. ISBN 9780191589676.

Bayes factor

- Typically you have hypotheses that involve **free parameters**
- To calculate the Bayes factor you have to integrate each $P(x \mid H_i)$ over the free parameters θ_i of the hypothesis H_i
- This integral will depend on the prior pdf for the free parameters $\pi_i(\theta_i \mid H_i)$

$$P(\mathbf{x} \mid H_i) = \int P(\mathbf{x}, \boldsymbol{\theta}_i \mid H_i) d\boldsymbol{\theta}_i = \int P(\mathbf{x} \mid \boldsymbol{\theta}_i, H_i) \pi_i(\boldsymbol{\theta}_i \mid H_i) d\boldsymbol{\theta}_i$$

- Note that this is often called the "**marginal likelihood**" or "evidence" of H_i
- The "marginal likelihood" $P(\boldsymbol{x} \mid H_i)$ depends on the prior of the free parameters $\boldsymbol{\theta}_i$
- To calculate the marginal likelihood one cannot use improper priors for $\pi(\boldsymbol{\theta}_i)$

Ockham's razor

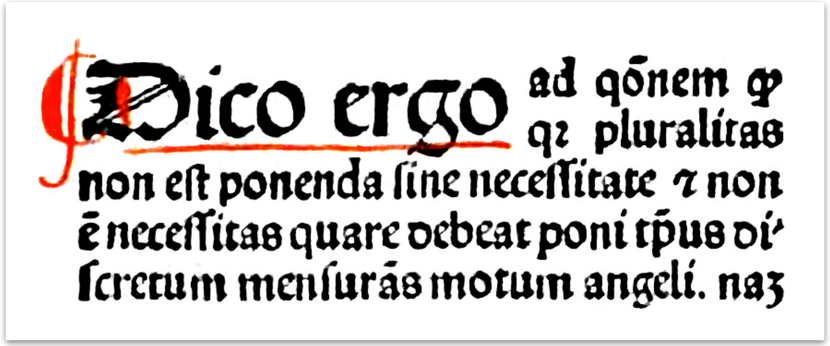
- The Bayes factor for H_i versus H_j is then given by the ratio of the likelihoods of the two hypotheses marginalised over the free parameters of each:

$$K = \frac{P(\mathbf{x}\,|\,H_i)}{P(\mathbf{x}\,|\,H_j)} = \frac{\int P(\mathbf{x}\,|\,\boldsymbol{\theta}_i, H_i)\pi_i(\boldsymbol{\theta}_i\,|\,H_i)d\boldsymbol{\theta}_i}{\int P(\mathbf{x}\,|\,\boldsymbol{\theta}_j, H_j)\pi_j(\boldsymbol{\theta}_j\,|\,H_j)d\boldsymbol{\theta}_j}$$

- Models with large ranges (volumes) of parameter space tend to have smaller marginal likelihoods

➡ Models with more adjustable parameters (volume) get smaller marginal likelihood and are therefore disfavoured

Bayes hypothesis testing has Ockham's razor baked in!


Dico ergo ad qōnem q
qr pluralitas
non eſt ponenda ſine neceſſitate ꝛ non
ē neceſſitas quare debeat poni tp̄us di'
ſcretum menſurās motum angeli. naz

Part of a page from John Duns Scotus's book *Commentaria oxoniensia ad IV libros magistri Sententiarus*, showing the words: "*Pluralitas non est ponenda sine necessitate*", i.e., "Plurality is not to be posited without necessity" Wikipedia.org





Jupyter notebook

Week4_BayesFactor.ipynb

Exercise for week 4

Week4_exercise.ipynb

